

Does Price Remember Where It Turned?

A location field for E-mini S&P 500 futures, tested against fake markets with no memory of price.

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INSTRUMENT	DEVELOPMENT	HELD OUT
E-mini S&P 500 futures (ES), one-minute bars	2022-01-03 to 2023-12-29	2024-01-02 to 2025-01-31
ZERO POINT		
19–20 sign-flipped copies of the real path		

Summary

Question. Traders see price stop at old highs and lows it has not traded through, and run through stretches where little has traded. Do large one-way moves in ES futures start, or resume after a pullback, more often at some prices than at others at the same time? And can those prices be read in advance from the price's own history?

Answer. Yes, by a few percentage points. Places that the field ranks above most of the places visited just before them are more often places from which a large move can be joined.

Why it could happen. A sharp turn leaves orders behind: the unfinished part of the order that turned price, stops placed just beyond it, and new entries that use it as a reference. They act again when price returns, whether it comes back to start a new move or as a pullback inside one, until price trades through them or they age out after a few weeks. Where little has traded, nothing absorbs a move once it has started.

The field. For any price and time, how many times more likely than at an ordinary place a large move is to start or resume there, if price comes. It is built from those two ingredients: big turns from one to four weeks earlier that price has not broken, and thin traded volume. It was fitted on 2022–2023 and frozen before the held-out data were read.

The yardstick. Every number is set beside the same pipeline run on fake markets, 19 in development and 20 held out: the real path cut into 13-minute blocks whose directions are flipped at random. The fakes keep volatility and its clustering but erase any memory of price, so what the market does and what the measurement does by itself can be told apart.

The result. In the held-out year, each place was ranked against the places of the previous trading day. Those in the top tenth went on to move eight times the session's typical swing 17.7% of the time, against 11.3% for the rest: a gap of 6.4 points, larger than in any of 20 fake markets. Taking the top fifth agrees; ranking against the previous week gives smaller gaps that beat 19 of the 20. It was a calm year with 2,927 places. In development the same rules gave gaps of 3.4 to 5.0 points, above every fake market, and the fit held when each design choice was moved one at a time.

Where the edge comes from. In development, about half of the edge beyond the fakes comes from pullbacks that hold inside a move, and about a third from the first such place of each session. At the exact origins of large moves the field does no better than on the fakes: origins sit at new highs and lows, where any field that favours thin ground ranks them high.

Tied to the price itself. When the field reads the history of a price 4σ or 8σ away, and the fake markets are read the same way, the edge disappears, at pullbacks and first key places alike.

What it does not show. One-minute bars carry no order book, so the results are the footprint the explanation predicts, not a direct view of the orders. The field does not say which way price will leave a place. The moves take days, not hours: the median time to eight times the swing is about 47 hours.

Why read it. Section 6 sets each prediction of the explanation beside what the data showed, including the explanations that failed, and shows where the edge comes from. Section 7 lists nine ways the measurement fooled itself, and how each was caught.

1 What traders see

Watch an intraday chart of ES long enough and some prices seem to matter more than others.

- Price stops, or turns sharply, at the price of an earlier sharp turn that it has not traded through since, often a high or a low from the previous week or two.
- Inside a trend, a pullback often stops at such a price, and the move resumes from there.
- Once price has traded through such a price, it stops mattering.
- Price crosses quickly through stretches where little has traded, and grinds inside ranges where a lot has traded.

Traders know these as support and resistance, and as the thin and thick parts of a volume profile. Mostly they are anecdotes. This note asks whether they hold up when measured: do large one-way moves start, or resume after a pullback, more often at such prices than at other prices at the same time, and can those prices be read in advance from the price's own history?

The study measures this as a relative rate. A place is compared with the places price visited just before it, over the previous day or week, and every difference is set beside what fake markets without memory produce. How such places might be traded is outside the scope of this note.

2 Why it might happen

Price moves because traders react to information. A chart cannot show the information, but it shows the footprint of the reaction. So each pattern above should be stated as a sentence about traders: who did what, what they left behind, and how they react when price comes back.

- **Orders left at a turn.** A sharp turn means someone absorbed the flow there: a large buyer at a low, a large seller at a high. If that order was not finished when price left, the rest still waits near the price and absorbs again when price returns, whether price comes back to start a new move or as a pullback inside one.
- **Stops and entries anchored to the turn.** Traders who entered on the turn, or who use it as a reference, place their stops just beyond it and their new entries near it. Orders therefore gather around the turn rather than exactly on it.
- **Orders have a lifetime.** A turn formed today is not yet a reference for many traders. Last week's highs and lows are the references most traders use. After a month or so, positions have been closed and orders cancelled.
- **Crossing uses them up.** Once price trades through the turn by a clear margin, the waiting orders are filled and the stops triggered, and nothing is left there.

- **Thin ground lets a move run.** Where little has traded, no dealer holds inventory to lean against a move and no earlier trades anchor the price, so once a disagreement there is settled, price can travel far. Inside heavily traded areas, dealers trade both ways and price grinds.
- **Orders change how far, not whether.** On any path, price turns a little all the time. Orders left at a price matter for whether a turn there becomes a large move.
- **Relative to the recent past.** How large a move must be to matter depends on who is active, and that changes from day to day. A place is therefore judged against the places price visited just before it.

This explanation makes predictions that can be checked:

1. Large moves start, or resume after a pullback, more often near recent turns that price has not broken than at other places at the same time. Turns from all of history do not help, because almost every price has turned at some point.
2. The effect is strongest for turns about one to two weeks old and fades within about a month.
3. It reaches a short distance around the turn's price, further on the side price is heading to.
4. A turn that price has traded through carries nothing.
5. Thin ground adds to the effect, and the inside of heavily traded areas starts fewer large moves.
6. The difference shows for large moves, not small ones.
7. None of it appears in fake markets without price memory, and it is tied to the price itself rather than to the market's general state.

The explanation came from watching charts and was sharpened while exploring the development sample. Results there describe the pattern; the held-out year tests it.

3 How it was tested

Data and units

- **Prices.** ES one-minute bars, stitched into one continuous front-month series without look-ahead. The contract for trading day D is the one that traded most on day $D-1$, rolls only go forward, and earlier prices are scaled by the ratio of the two contracts' closes. Trading days run from 18:00 to 17:00 New York time.
- **Scale.** All distances are in units of $\sigma = 0.206\%$ of price, about 8 points with ES at 4,000.
- **Samples.** Development: 2022-01-03 to 2023-12-29, with bars from 2021-05-17 as warm-up. Held out: visits from 2024-01-02 to 2025-01-31, with price histories from 2023-07-01 and outcomes read to 2025-03-31. Data from April 2025 onward are kept back for later tests.

Definitions

Turn and its level

Price is read as a zigzag at σ : a turn is confirmed once price reverses by σ from an extreme. A turn's level is the smaller of the swing into it and the swing out of it, so a pullback inside a long move keeps the size of the pullback. A line at a turn's price is broken only when price trades more than 1σ beyond it.

Session level

The typical swing of each trading day and session: the median swing between consecutive turns, for the regular session (09:30–16:00 New York time) and for the rest. Features use the level known at the moment: the median of the session's finished swings once there are five, otherwise the previous same session's level. The median session level is 1.75σ , about 14 ES points, so eight times is roughly 115 points.

Zone, visit and place

Zones are $\pm\frac{1}{2}\sigma$ wide and centred every $\frac{1}{2}\sigma$. A visit begins on the first bar that reaches the zone and ends when a bar lies entirely 1.5σ from the zone's centre. Back and forth inside the zone is one visit, and one visit is one observation. A place is a visit to one of a set of non-overlapping 1σ zones, so each turn falls in exactly one place.

Outcome

For each place, the largest *move-out* among the turns formed in the zone during the visit: how far price went from the turn before trading 1σ back beyond it, in units of the session level. It is the most a trade entered at the turn, with a 1σ stop, could have gained. There is no time limit; a move not yet finished when the data end counts as a lower bound.

Key place

A place whose outcome reaches four times the session level (24% of places) or eight times (13%). Its turn may start a new move or end a pullback inside one; either way, a trade entered at the turn with a stop just beyond it could have caught a large move. Twice was too lenient: it marks 44% of places on the real path and on the fakes alike.

The outcome has no direction. A reversal counts through its turn. A pause followed by a break-through also counts, through the pullback turn it leaves in the zone. Price that runs straight through leaves no turn, so no move started there. Which way price leaves is a separate question.

Order of work

The key places were picked first, by what the moves from them did. Their surroundings at the moment price arrived were then described and set against the predictions of section 2. Only then was a field fitted on the development sample, frozen, and tested on the held-out year.

Fake markets as the zero point

Some differences between key places and other places appear whatever the market does, because features and outcomes are measured on the same path. To measure that part, the whole pipeline also runs on fake markets:

- the real one-minute path is cut into 13-minute blocks, the median duration of a leg at σ ;
- each block's returns are flipped in sign at random;
- the blocks are stitched back together.

Volatility, its clustering, the intraday pattern of activity and the path inside each block stay as they were. Prices end up at other levels, so what happened at a price before says nothing about what happens there now. The fakes go through the same feature search and fitting as the real path, so the zero point also absorbs the optimism of the search. There are 19 fake paths on the development sample and 20 on the held-out sample. Blocks of 7 and 26 minutes were also checked.

Only what was known at the time

Every reading uses only what was known on the bar the visit began: confirmed turns, printed bars, and the session level known then. This was also checked mechanically. For six places drawn at random, the data were cut after the visit's first bar and everything was rebuilt from scratch. Every input the field uses came out identical to the full-data run, as did all 560,000 rows of big turns around those places. In addition, on paths flipped bar by bar, which have no memory at all, every group of places must average within $1 \pm 3\%$ of its expectation; the outcome passes with deviations of 0.9–1.8%.

Comparing with the recent past

The field is relative by definition: a place is more or less likely than other places at about the same time. Each place is therefore ranked against the places price visited just before it, over the previous day or week. This cancels the overall activity of the period and the size of the players active in it, and uses nothing that came later.

4 The field

The development sample has 12,077 places; 24.4% are key at four times the session level and 13.1% at eight times. Of the readings suggested by section 2, two carried the effect in both development years, above all the fake markets.

- **Standing big turns.** Turns that were large for their session, formed one to four weeks earlier and not broken since. Their influence reaches a short distance around their price, leaning ahead of the approaching price, and is strongest for turns one to two weeks old, perhaps the previous week's highs and lows. Turns that price has already broken carry nothing.
- **Thin turnover.** Little traded volume in the zone and on both sides.

The field is a log-linear model of the chance of an eight-times move, fitted by Poisson maximum likelihood. The primary version, V2, multiplies two terms: one for standing big turns, weighted by distance and age through a kernel chosen from a grid of candidates, and one for thin turnover. The field is flat over most places and rises near standing big turns.

TABLE 1 Development sample, eight times the session level. In grey: median and maximum over 19 fake markets, each searched and fitted the same way.

VERSION	LIKELIHOOD GAIN PER 1,000 PLACES	TOP 10% VS BASE, POINTS	TOP 20% VS BASE, POINTS
V1 · turns	1.80 (0.27, 0.66)	+5.4 (+0.9, +3.9)	+2.2 (+0.5, +2.4)
V2 · turns × thin	2.38 (0.35, 0.67)	+5.1 (+1.1, +3.6)	+3.2 (+1.1, +2.2)
V3 · + old traded area, range edge	3.26 (1.05, 2.35)	+6.6 (+2.9, +4.8)	+4.4 (+2.4, +3.6)
V4 · + larger leg, trapped positions	3.49 (1.07, 2.47)	+6.9 (+3.1, +4.9)	+4.7 (+2.5, +3.7)
V2 with turn ages weighted equally	1.23 (0.25, 0.60)	+5.5 (+1.3, +3.6)	+2.8 (+1.1, +2.4)

Every version beats every fake market in likelihood gain and in its top tenth. Richer versions also manufacture more on the fakes, so net of the fakes they are about equal, and V2, with the fewest parameters, is the primary version. Weighting turn ages equally keeps about half the gain. So the result does not rest only on the age bands, which were chosen after seeing the development data.

Robustness. Each design choice was moved one at a time: the thin cut, the widths of two terms, the big-turn threshold, and the multiple at which the kernel is chosen. The real and fake paths were refitted from scratch each time, and V2's likelihood gain stayed above every fake market in all eleven settings. Fake markets built from 7- and 26-minute blocks give the same picture: V2's gain of 2.38 lies above all of them, with fake medians of 0.42, 0.35 and 0.31 for 7-, 13- and 26-minute blocks.

5 Held-out test

The field's parameters were written to a single file, and its SHA-256 fingerprint was recorded before the held-out data were read. Nothing was refitted, the held-out sample was never used to fit or choose anything about the field, and each fake market was scored with the same frozen field.

Each place is ranked only against the places that came before it: those of the previous trading day, or of the previous five, and the earlier places of its own day. If its field value is in the top 10% (or 20%) of that window, it joins the top group; places tied at the cut count in proportion. The top group is then compared with the rest on the chance of moving out eight times the session level. The ranking only looks back, so it can be applied as price arrives.

FIGURE 1 Held-out year. Gap in the eight-times rate between the top group and the rest, each place ranked against the places before it.

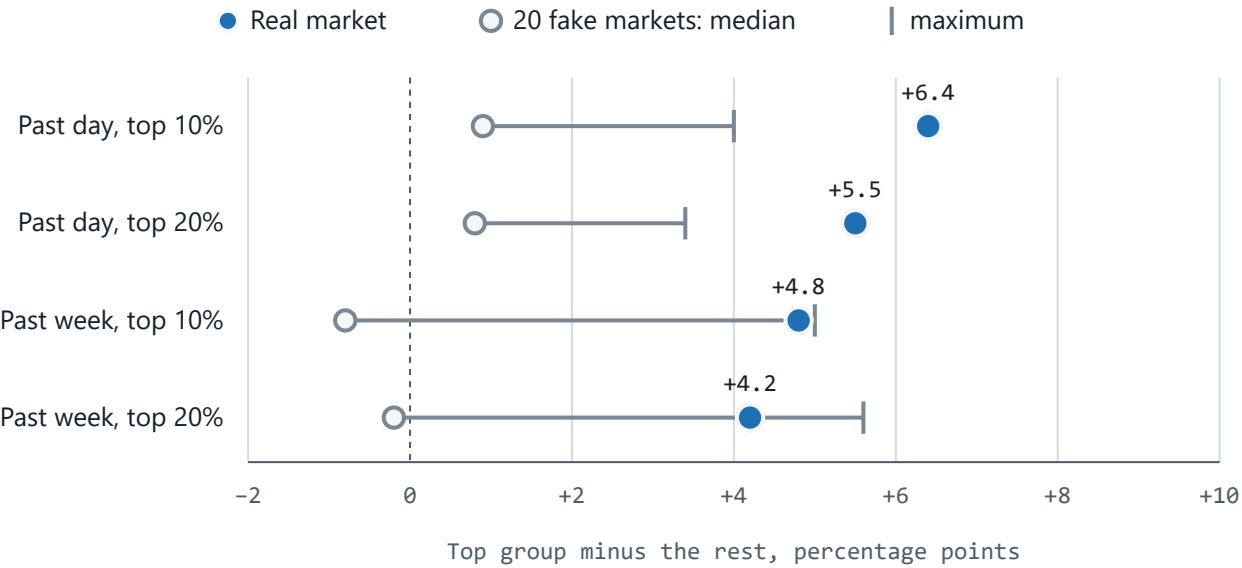


TABLE 2 Held-out year, each place ranked against the places before it. Eight-times rate of the top group and of the rest.

RANKED AGAINST	TOP GROUP	REST	GAP, POINTS	FAKES: MEDIAN / MAX	FAKES AT OR ABOVE
Past day, top 10%	17.7%	11.3%	+6.4	+0.9 / +4.0	0 of 20
Past day, top 20%	16.4%	10.8%	+5.5	+0.8 / +3.4	0 of 20
Past week, top 10%	16.3%	11.4%	+4.8	-0.8 / +5.0	1 of 20
Past week, top 20%	15.4%	11.2%	+4.2	-0.2 / +5.6	1 of 20

Ranked against the previous day, the top tenth went on to move eight times the session level 17.7% of the time, against 11.3% for the rest: a gap of 6.4 points, larger than in any of the 20 fake markets. The top fifth also beats every fake. Ranked against the previous week, the gaps are smaller, 4.8 and 4.2 points, and one of the 20 fakes reaches each of them. On the development sample the same four rules give +3.4, +3.5, +5.0 and +4.6 points, all above every fake market, whose medians sit near zero.

The held-out year was calm: 21 visits a day against 47 in development, giving 2,927 places and 350 places that moved out eight times.

How much evidence. The 350 key places are not 350 independent events. Of the 274 trading days with a full window, 69 contribute a positive excess to the past-day top tenth, and the five strongest carry a little over a quarter of it. A day-block bootstrap of that gap (1,000 draws) puts the middle 90% at +3.1 to +9.9 points, so the gap is not noise around zero. Against that, 20 fake markets resolve chance only to about one in 21 per rule, and the four rules are correlated.

Ranking within the whole period. As a check on how well the field orders places, each place was also ranked against all places of its day or week. This uses the field values of places that came later in the period, though none of their outcomes, so it is an evaluation rather than a rule one could trade. Within each day, the top tenth went on to move eight times 19.8% of the time, against 11.1% for the rest. All four daily and weekly groupings beat all 20 fakes (Table 3). At the finest grain, trading day × session, where a group holds about six places, the top fifth still beats every fake market and the top tenth all but one.

TABLE 3 Held-out year, each place ranked against all places of its day or week; an evaluation of the field's ordering, not a tradable rule. Eight-times rate of the top group and of the rest of its period.

GROUPING	TOP GROUP	REST	GAP, POINTS	FAKES: MEDIAN / MAX	FAKES AT OR ABOVE
Daily, top 10%	19.8%	11.1%	+8.7	+2.1 / +7.1	0 of 20
Daily, top 20%	16.7%	10.8%	+6.0	+1.4 / +4.2	0 of 20
Weekly, top 10%	18.4%	11.2%	+7.2	−0.4 / +7.0	0 of 20
Weekly, top 20%	16.5%	10.8%	+5.7	−0.3 / +4.2	0 of 20
Day × session, top 10%	—	—	+5.0	+2.4 / +5.5	1 of 20
Day × session, top 20%	—	—	+4.4	+1.4 / +4.2	0 of 20

Pooling. Scored across all thirteen months at once, the top tenth still beats the rest by 3.4 points (15.0% against 11.6%), but one of the 20 fake markets does better (4.4). Pooling compares calm periods with active ones and does not match the field's definition; the drop in the gap shows how much of the edge is relative to its period.

6 What was verified, and what was not

Predictions against results

TABLE 4 The predictions of section 2 beside the data. Development results describe the pattern; the held-out year tests the field that combines predictions 1 to 5.

PREDICTION	WHAT THE DATA SHOWED	STATUS
1. Near recent turns that price has not broken, large moves start or resume more often; turns from all of history do not help	A standing big turn from the last four weeks raised the eight-times rate by 2.6–3.0 points, above all 19 fakes. Counting every earlier turn showed no difference.	Supported in development, and held out through the field
2. Strongest for turns one to two weeks old, fading within about a month	Strongest for turns one to two weeks old; little from turns one to three months old.	Supported in development
3. Reaches a short distance around the turn, further on the side price is heading to	Strongest in the turn's own zone, reaching further ahead than behind.	Supported in development
4. A turn that price has traded through carries nothing	Broken turns added nothing. Turns broken within the last day went with fewer large moves, below every fake.	Supported in development
5. Thin ground adds; heavily traded ground damps	Thin zones raised the rate by 2.0 points, above all fakes. Fewer moves started inside heavily traded areas, but within the fakes' range.	Thin: supported. Damping: same direction, not beyond the fakes
6. The difference shows for large moves only	No difference at twice the session level; the ratio grows toward eight and sixteen times.	Supported in development
7. Nothing without price memory, and tied to the price itself	The fakes show small gaps throughout. Read from a price 4σ or 8σ away, with the fakes read the same way, the edge disappears.	Supported in development

Explanations that did not hold

Not every explanation survived. Trapped positions, held by traders who bought at a heavily traded band and want out when price returns, showed no effect of their own once the other terms were held. The way price arrived made no difference, whether measured

by its speed, its volume, or its direction relative to the larger move. Traces of swept stops and of urgent buyers or sellers pointed the expected way but stayed inside the fakes' range. Richer versions of the field, which add the old traded area, the edge of the recent range, the larger leg and trapped positions, gained nothing once the fakes' own fitting gains were subtracted (Table 1).

Where the edge comes from

A key place can be where a large move begins or where a pullback inside it holds. In both cases a trade entered at the turn, with its stop just beyond, could have caught a large move. The key places therefore split into three kinds:

- **Origins.** Turns that are large in their own right: both the swing into the turn and the swing out of it reach eight times the session level.
- **The first key place of a session.** The first place in a trading day and session from which a large move followed, when its turn is not an origin. Often the move began in an earlier session and this is its first pullback that holds.
- **Later key places.** The others in the same session, mostly pullbacks that hold inside a move already under way.

FIGURE 2 Schematic, not data. The origin is a large turn in its own right. Each pullback that holds is a smaller turn from which the rest of the move could still be caught.

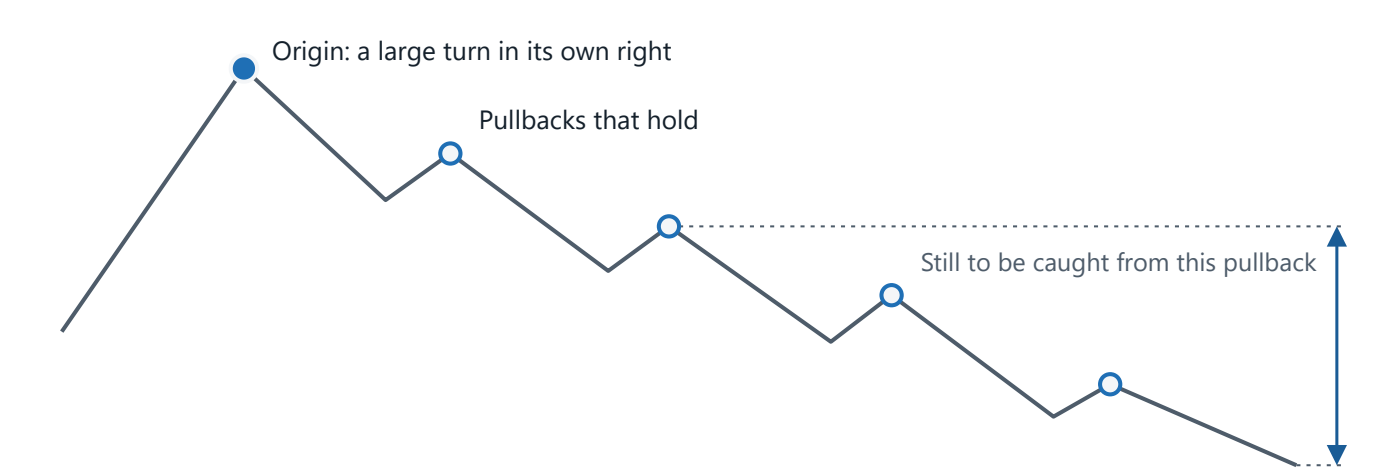


TABLE 5 Development sample, eight times the session level. The gap between each day's top tenth by field and the rest of that day, ranked within the whole day as in Table 3, split by kind of key place. The fake markets are scored with the same frozen field.

KIND	KEY PLACES	IN THE TOP TENTH	IN THE REST	GAP, POINTS	FAKES: MEDIAN (MAX)	BEYOND THE FAKES' MEDIAN
Origins	98	3.1%	0.6%	+2.5	+2.2 (+3.0)	+0.3
First key place of a session	480	6.8%	3.7%	+3.1	+1.9 (+3.0)	+1.3
Later key places	1,005	8.1%	8.4%	−0.3	−2.2 (−1.0)	+1.9
All key places	1,583	18.0%	12.6%	+5.4	+1.7 (+4.5)	+3.7

- **Origins: nothing beyond the fakes.** The field puts origins in its top tenth five times as often as in the rest, but it does the same on the fake markets. Origins sit at the session's highest or lowest zone almost by construction (99%, against about 20% of other places), and price has hardly traded there, so they read as thin (90%, against about 30%). Ranked only among a session's new highs and lows, origins sit at 0.71, against 0.73 on the fakes.

- **Pullbacks: the largest part of the edge.** On the fake markets, the field's top tenth holds clearly fewer pullbacks than the rest of the day (−2.2 points): without memory, pullbacks rarely sit where the field is high. On the real path they sit there as often as anywhere else, 1.9 points beyond the fakes and above all 19.
- **The first key place of a session:** 1.3 points beyond the fakes, above all 19.

Ranking within each day and session instead gives the same picture: origins −0.2 beyond the fakes, first key places +1.0, later key places +1.3. So the field's contribution beyond chance is about where a large move can be joined, mostly at the session's first such place and at pullbacks that hold, not about the exact turning point where the move begins.

Memory of place, or a general state?

A field that only says "price is somewhere new, in a trend" would also beat the fake markets, which have no trends longer than 13 minutes. To separate the two, the field was read for each place from the history of a price 2, 4 or 8σ away, keeping the place's time, direction of approach and session level, and the whole-period comparison of Table 3 was repeated on the development sample. The fake markets were read with the same shifts, because a shifted reading also changes what the field picks up mechanically from where the place sits.

FIGURE 3 Development sample. Gap between the top group and the rest of its period, averaged over the four whole-period groupings (daily and weekly, top tenth and top fifth), when the field reads the history of a shifted price. The fake markets are read with the same shift.

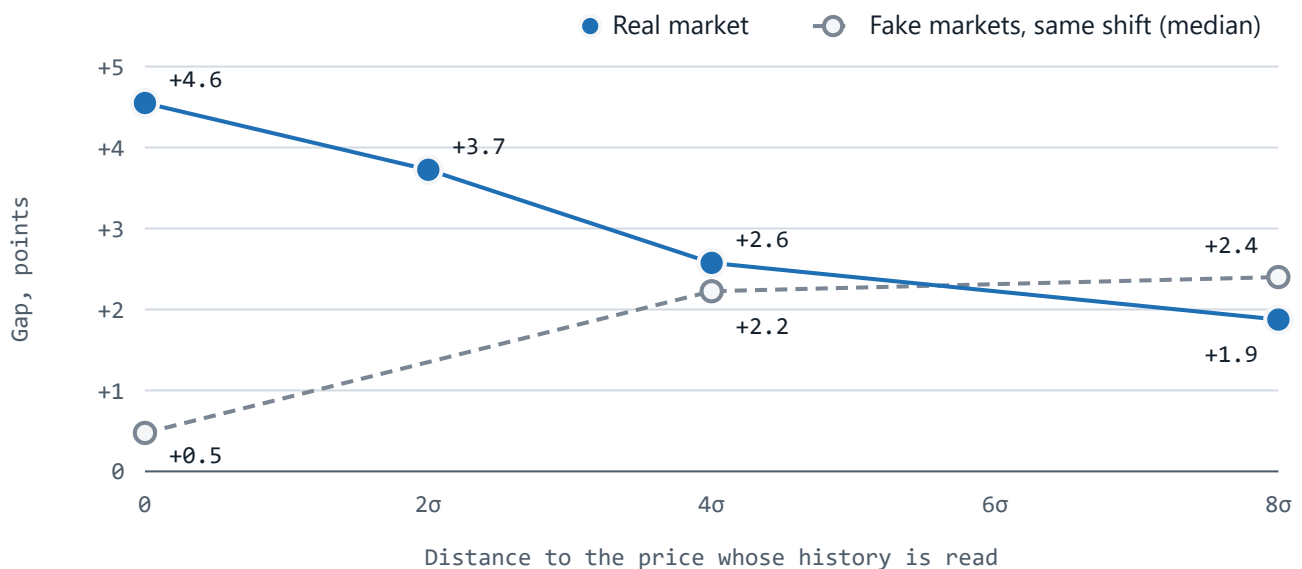


TABLE 6 Development sample. Gap beyond the fakes' median, in points, when the field reads the history of a price shifted up and down by the given distance and the 19 fakes are read the same way; averaged over the four whole-period groupings. In brackets: how many fakes reach the real gap.

KIND	OWN PRICE	4Σ AWAY	8Σ AWAY
Later key places (pullbacks)	+2.1 (0)	+0.2 (5)	−0.5 (19)
First key place of a session	+1.5 (0)	−0.1 (11)	−0.1 (12)
Origins	+0.4 (2)	+0.1 (5)	+0.2 (4)
All key places	+4.1 (0)	+0.4 (5)	−0.5 (17)

Beyond the fakes read the same way, the edge is +4.1 points at the place's own price, +0.4 at 4σ and below zero at 8σ. It disappears at the pullbacks and at the first key places alike. Each of the field's two terms, ranked alone, loses its edge the same way: the thinness that counts is the place's own, and so are the turns, apart from a small part that still shows 4σ away. So the edge belongs to the history of the price itself and its immediate surroundings, not to a general state of the market. Without the same shift on the fakes the

comparison would mislead: read 8σ away, the fake markets themselves show a gap of +2.4 points. The check ran on one path and its fakes, in the development sample.

Time scale

The moves take days. From the turn, the median time to reach 2, 4 and 8 times the session level is 1.4, 7.3 and 47 hours, and only 28% of eight-times moves get there within 23 hours. In the first hour the field's top tenth moves less than the rest: it reaches twice the session level 12.6% of the time, against 19.0%. Its advantage appears after days: within five days it reaches four times 4.7 points more often. For intraday trading, the field on its own is background.

What these data cannot show

The explanation in section 2 is about orders and positions, and none of them is visible in one-minute bars: the data have no order book and no record of who bought or sold. The results match the footprint the explanation predicts; they do not show the orders themselves. Nor can they show which explanation does the work, since unfinished orders and stops beyond a turn leave the same footprint.

7 Issues found and fixed

Much of the effort went into finding ways the measurement could fool itself. These are the ones that mattered.

ISSUE	HOW IT SHOWED UP	EFFECT	FIX
Session level read ahead within the day	Comparing the code with the written definitions	The big-turn threshold used the whole session's level. About 9% of big turns changed class, and turn effects shrank by a tenth to a fifth.	Use the level known when the visit begins
Cuts ranked within the year	A full audit of "as known then" rules	Thresholds used later data from the same year. V2's gain moved from 2.43 to 2.38.	Freeze every cut on the development sample
Approach measured from an unconfirmed turn	A review of older readings	For 23.2% of places the starting turn was not yet confirmed. Only exploratory tables used it.	Start from the last confirmed turn; add the truncation test of section 3
Ties broken by storage order	The field takes few distinct values	The top fifth took 1,131 of 3,055 equal values in table order, all from 2022: +4.0 points instead of +3.2.	Count tied places at the cut in proportion
A ratio above its bound	An efficiency above one for 19.4% of places	It read the turn bar's wick. Corrected, its top tenth only marked visits right after a fresh turn.	Drop the term
Rows per touch instead of per visit	One move produced 10–20 rows	The evidence was overstated: 617 rows were 30 visits on 6 trading days.	One visit, one observation
Differences that every market shows	The fake markets	The approaching leg's length differed by -0.133 on the real path and by -0.136 to -0.164 on the fakes.	Report every number against the fakes

ISSUE	HOW IT SHOWED UP	EFFECT	FIX
A zero point read differently from the real path	Shifted readings of the real path were first compared with unshifted fakes	A third of the edge seemed to survive a shift of 8σ . Fakes read with the same shift reach that level too, so that third was mechanical.	Read the fakes exactly as the real path is read
A pooled comparison for a relative field	Scoring all thirteen held-out months together	Calm and active months are mixed; the gap shrinks to 3.4 points, within one fake market's reach (section 5).	Rank each place against places of about the same time

8 Limitations and next steps

- **One instrument and one held-out year.** The held-out year was calm, with 2,927 places and 350 eight-times key places.
- **Few independent moves.** Large moves cluster. In development, the top tenth's 220 key places come from 36 separate moves, and the five strongest sessions carry 42% of its excess. Evidence is better counted in moves than in places.
- **Coarse probabilities.** Twenty fake markets resolve probabilities only to about one in 21.
- **Origins not singled out.** At the exact origins of large moves the field does no better than on fake markets; its edge lies at pullbacks that hold and at the first such place of a session. That breakdown was measured on the development sample only (section 6).
- **Multi-day horizon.** On its own, the field is background for intraday trading.
- **A key rate is not a win rate.** The note measures where large moves start or resume, not how to enter, where to stop or when to exit.

Next. Extend the study to several scales, and test it on data not yet used, from April 2025 on, and on other instruments.

Validation checklist

CONCERN	WHAT WAS DONE
Explanation	Every ingredient is tied to a stated trader behaviour with a prediction, and the explanations that failed are reported (sections 2 and 6)
Look-ahead	Every reading uses only what was known on the visit's first bar; checked mechanically by cutting the data after that bar and rebuilding six places from scratch (section 3)
Null model	Sign-flipped fake markets keep volatility, its clustering, the time-of-day pattern and the path inside each 13-minute block; they destroy price memory and any direction beyond 13 minutes. Blocks of 7 and 26 minutes agree
Placebo	The field read from prices 4σ and 8σ away, with the fakes read the same way; the edge disappears (section 6)
Mechanical differences	Every number is reported beside the same pipeline on the fakes; differences the fakes reproduce are discarded (section 7)
Selection in the search	82 feature rows per threshold against 19 fakes: about one row in ten lies outside their range by chance. The kernel was chosen from 2,304 candidates, and the fakes searched the same grid, so the zero point absorbs the optimism of the search
Frozen parameters	One file, SHA-256 recorded before the held-out data were read; nothing refitted
Sensitivity	Each design choice moved one at a time, real and fake paths refitted; block length varied (section 4)
Calibration	On paths flipped bar by bar, with no memory at all, the outcome is flat to within $1 \pm 3\%$ (section 3)
Evidence counting	Visits, not touches; moves and days reported beside places; day-block bootstrap of the held-out gap (sections 5 and 8)
Resolution of the test	20 fakes resolve chance to about one in 21 per rule; the rules are correlated (section 5)

Reproducibility

- Python research code. Market logic runs bar by bar, as it would live; batch runs only replay the bars.
- 268 unit tests at the last full run.
- Every number has an entry in a parameter registry, with its origin. The field's parameters are frozen in one file whose SHA-256 begins 36723a79.
- A detailed internal report records every step, including the dead ends.

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